APPENDIX 6

A Brief Guide to Writing in Chemistry

To avoid criticism, do nothing, say nothing, be nothing (Elbert Hubbard)

This document is a guide to assist students in chemistry courses with writing and formatting laboratory reports and research reports. An important goal of an undergraduate curriculum is for our students to organize and communicate research results effectively and to write with acceptable scientific style. By providing many of the common stylistic, grammatical and organizational points, this document provides students with a reference to use throughout their study of chemistry. However, an effective way to improve scientific writing is to read primary research papers. Students are encouraged to consult papers in leading chemistry journals such as *The Journal of the American Chemical Society* in order to appreciate the formatting and writing style of research reports in the field of chemistry.

Formatting a Report

Layout Use 12-point Times New Roman font and double spacing to allow space for comments and corrections. Number all pages, including those in appendices.

Organization A standard lab report or research paper should be formatted with sections.

1. *Title*—give the title of your experiment or research report a meaningful name.

poor: "The Synthesis of a Coordination Compound"
better: "The Synthesis of a *cis*-Platin"

The title is followed by your name and the date submitted. If you worked with partners, list their names next to yours, but put an asterisk after your name to indicate that you wrote the report.

2. *Abstract*—the abstract should be able to "stand alone." This means that someone should be able to read *only* the abstract and understand the basic nature of your report. For this reason, a good abstract clearly identifies the purpose of the

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experiment *and the important results; a good abstract contains a summary of your results*. Avoid pedagogical comments such as "this experiment helped us learn about the nature of chemical kinetics" or "the goal of this experiment was to learn about metal ions." Although those are important aspects and goals of a lab experience, the lab/research report should focus on the data and results. Avoid starting your abstract with "The purpose of this experiment was ..." Background information on the theory or applications of your experiment belongs in the Introduction section. Avoid referencing other parts of the report, because the abstract should be able to "stand alone." Be specific about what was done: name the reagents or types (not models) of instruments that were used, identify the products of a reaction, and indicate numerical values that were measured or calculated, etc. Avoid vague statements such as "a metal complex was prepared and the percentage yield was calculated." A better abstract would read "hexaammine cobalt(III) chloride was prepared from cobalt (II) chloride, ammonia, ammonium chloride, and hydrogen peroxide. The yield was 8.45 g (64% based on cobalt)." *Tip*: when writing a full report, write the abstract last.

3. *Introduction*—the Introduction section explains to the reader what basic scientific question is being addressed. This can be offered through general background material or a brief historical perspective on the topic being investigated followed by brief summaries, with references, of previous related work. An effective introduction guides the reader from the larger research area, through specific examples of progress in the field to a clear statement of the research problem or approach being addressed in the current report.
4. *Experimental*—this section includes a description of your experimental procedure, and names of instruments used. For lab courses, the procedure can simply *reference the lab manual*, listing any changes to the published procedure. *Do not rewrite the lab manual*. For advanced labs or independent research projects, the experimental section should provide all the necessary detail for someone to be able to reproduce your work. Often, an Experimental section is subdivided into Materials (sources and purity of reagents used), Preparation of Compounds (with procedure, and summary of characterization by NMR, IR, UV–vis spectroscopies, melting point, chromatography, or elemental analysis) and Instrumentation (manufacturer, description of any adaptation or sample preparation) sections. Consult chemistry journals to see examples of Experimental sections for various types of reports. *Tip*: a good experimental section should allow another person, using what you have written and a lab manual, to completely reproduce what you did in the lab.
5. *Results and Discussion* (may be single or separate sections)—the Results section is a summary of your experimental results, important observations and numerical data (preferably in tabular form, when possible). Avoid including especially long tables of raw data, instead presenting the results of calculations or averaged values. A sample calculation may be included in this section or in an Appendix, and a description of equations used in your calculations should be presented. Hand-written calculations may be acceptable for lab reports.

   A Discussion section should take the form of an analysis of your results. Comment on the purpose of the experiment. What do the results indicate? What are sources of error (experimental uncertainty/precision)? What additional

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experiments could help address any dangling ends? Do the results agree with what others have found? Do the results support a model or hypothesis? For some lab courses, you can use this section to answer any questions presented in the manual or in class. Although you should answer the questions in the lab manual, this section should have the style of flowing prose, not simply answers to numbered questions.

6. *Conclusion*—summarize your results and discussion with a short conclusion that is more than simply a reiteration of your results. Phrase it in terms of the broader questions addressed in the Introduction.
7. *References*—citations of the literature used in the previous sections (see below).
8. *Appendix*—graphics may appear here, along with lengthy calculations or additional material not needed by the reader when reading the report.
9. *Graphics*—graphics include tables, figures, schemes and chemical structures. Tables are columns of measured and/or calculated values or observations. All quantities should have units and be expressed using proper significant figures and scientific notation. Important experimental conditions should be listed as footnotes, especially when the table includes data obtained under different experimental conditions. Figures include: spectra, graphs, cartoons of experimental set-up or other drawings intended to show an *object*. Schemes include: reaction mechanisms, experimental flow charts or other drawings that are intended to show a *process*. All tables, figures and schemes should be numbered sequentially and must be mentioned in the text. Graphics are best when large, and should be at least half-page in size; they can be inserted into the text or included at the end of the manuscript in the Appendix. Chemical structures can appear in the text and should be labeled with the same name, formula or compound number that appears in the text.

Sentence Structure and Writing Style

1. *Beginning a sentence*. Avoid beginning a sentence with a symbol, numeric value or equation.

   incorrect: 315.6 mg of ammonium chloride was added to the solution, which was then heated to 50°C.

   correct: After the addition of 315.6 mg of ammonium chloride, the solution was heated to 50°C.

   incorrect: v is both the vibrational frequency and the IR radiation frequency.

   correct: The frequency v refers to both the vibrational frequency and the frequency of IR radiation.

2. *Dangling modifiers and illogical construction*—check that a modifier phrase or the pronoun "it" actually refers to the intended subject (see also: Subject–verb agreement).

   incorrect: Being slightly wet, I first dried the solid in a dessicator.
   Was I wet or was the solid?

   correct: Because the solid was slightly wet, it was dried in a dessicator.

   incorrect: After transferring to a larger flask, the solution was heated to a boil.
   Did the solution transfer itself?

   correct: The solution was transferred to a larger flask and heated to a boil.

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incorrect: A diagram of the SARS virus is now available. We obtained it from an online source. *The SARS virus is available online?*

correct: A diagram of the SARS virus is now available from an online source.

3. *Equations*—equations typically appear as a separate line from the text and are numbered sequentially throughout the manuscript. Equations can then be referred to by number.

   Example: The quenching rate constant can be calculated using the Stern–Volmer equation:

$$\Phi_0/\Phi_q = 1 + k_q \tau_0 [Q] \tag{A.6.2}$$

4. *Spaces*—there should be a space between a quantity and its units and between a quantity or word and subsequent parenthetical phrase.

   Examples:

   6.626 J s

   25.15 K = 298.15 °C

   45 ml

   456 nm (34,000 M^{-1} cm^{-1})

5. *Personal pronouns*—by tradition, scientists tend to avoid using the personal pronouns "I" and "we" and "you" in most technical communications. The use of third person instead of first person is standard for research results (see also: Active voice).

   first person: I heated the solution at 100 °C for 1 h and I noticed that it turned blue.

   third person: When heated at 100 °C for 1 h, the solution turned blue.

6. *Pedagogical comments*—avoid including pedagogical comments in a report or scientific communication. Phrases such as "this experiment helped us learn about ligand field theory" or "the goal of this experiment was to learn about ligand substitution kinetics" are addressing the process of learning not the science of the experiment. Although those *are* important aspects and goals of the lab experience, the lab report should focus on the data and results.

7. *Personification*—molecules and equipment are not people, so do not personify them in your writing.

   incorrect: Potassium nitrate really wants to dissolve in water.

   correct: Potassium nitrate is very soluble in water.

   incorrect: Sodium wants to lose one electron to form Na^+.

   correct: Oxidation of Na to Na^+ is thermodynamically favorable.

   incorrect: The spectrum shows two bands of equal intensity

   correct: Two bands of equal intensity appear in the spectrum.

8. *Plural nouns*—"Data" is plural for "datum," "spectra" is plural for "spectrum," "phenomena" is plural for "phenomenon," and "formulae" is plural for "formula." The amount of chemical reagent is singular, so use the correct verb tense.

   incorrect: Data was acquired and a spectra is in the appendix.

   correct: Data were acquired and a spectrum is in the appendix.

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incorrect: While the solution boiled, 5.0 g of KBr were added.

correct: While the solution boiled, 5.0 g of KBr was added.

9. *Prepositions*—Do not forget "of" between quantities and substance name.

   incorrect: . . . and 10 ml MeOH was added.

   correct: . . . and 10 ml of MeOH was added.

10. *Redundant or unnecessary phrases*

    incorrect: A photon of light having a wavelength of 530 nm . . . *if not "of light," what was the photon made of?*

    correct: Light having a wavelength of 530 nm . . .

    incorrect: In this experiment, *cis*-platin was prepared from tetrachloroplatinate. *If not this experiment, then in which experiment?*

    correct: *cis*-Platin was prepared from tetrachloroplatinate.

11. *Subject–verb agreement*—are you stating that an inanimate object is drawing a conclusion, or suggesting a strange cause and effect? (See also: Dangling modifiers.)

    incorrect: The IR spectrum implies that water is in the sample of $Cr(phen)_3^{3+}$. (*spectra do not imply, people do*)

    correct: The presence of water in the sample of $Cr(phen)_3^{3+}$ is inferred from the IR spectrum. (*the presence is inferred*)

    incorrect: Water was present in the product because of the peak at 3200 cm^{-1} in the IR spectrum. (*the peak in the spectrum did not cause water to be present*)

    correct: The peak at 3200 cm^{-1} in the IR spectrum indicates that water was present in the product. (*water caused the peak in the spectrum*)

Verbs

1. *Active voice*—by avoiding personal pronouns, scientists often depend excessively on the passive voice, which can weaken the writing style. *When possible*, replace passive voice with active voice.

   passive voice: A color change was observed when the solution was heated.

   active voice: A blue color formed as the hot solution.

   passive voice: There was some solid $CrCl_3$ that did not dissolve.

   active voice: Some $CrCl_3$ solid did not dissolve.

2. *Subject–verb agreement*—based on whether the subject is singular or plural, use the correct verb tense. A quantity used is a singular subject, even when that quantity is in a plural form of units.

   incorrect: 45 mg **were** added

   correct: 45 mg **was** added

3. *Verb tense*—past tense is used to describe a procedure that you followed in an experiment. Present tense is used to describe a scientific fact, such as the properties of a molecule.

   Examples: Potassium tetrachloroplatinate was used as the starting material for the synthesis of *cis*-Platin. When dissolved in water, K_2PtCl_4 forms a cherry-red solution.

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4. *"Verbing" a noun*—do not turn nouns into verbs.

   incorrect: ammonia complexes to cobalt ions
   correct: ammonia forms complexes with cobalt ions.

   incorrect: the mixture was centrifuged to separate the solid.
   correct: The solid was separated from the mixture using a centrifuge.

   incorrect: The solution was rotovapped to dryness
   correct: The solvent was removed by rotary evaporation

Abbreviations, Formulae, and Numerals

1. *Standard abbreviations*—use standard *JACS* abbreviations (*note*, not all journals use exactly the same abbreviations):

   Examples: ml = milliliter; μg = microgram; nM = nanomolar
   h = hour; min = minute; s = second
   K = degrees Kelvin, °C = degrees Celsius

2. *Chemical formulae*—use subscripts, superscripts, parentheses, and symbols appropriately in chemical formulae.

   Examples: $Cr^{3+}(aq)$
   $K_2[PtCl_4]$
   $[Ru(bpy)_3^{2+}](PF_6)_2$

3. *Compound numbers*—compounds can be numbered if repeated long compounds names become cumbersome. The number should be defined (usually in bold or underlined) somewhere early in the manuscript, often when it is first presented.

   Example: "Investigations into the fluorescence of 8-hydroxyquinoline (**1**), 4-iodo-8-hydroxyquinoline (**2**) and 2-methyl-4-iodo-8-hydroxyquinoline (**3**) are described in this paper. Recrystallization of **1** and **2** afforded analytically pure samples, but vacuum sublimation of the methyl derivative (**3**) was necessary to remove fluorescent impurities."

4. *Decimal places*—for values less than unity, use a leading zero. Avoid writing values having too many zeroes; use scientific notation.

   Examples: "0.15 μL" not ".15 μl"
   "2.3×10^{-5} M" not "0.000024 M"

5. *Defining abbreviations*—abbreviations for chemical compounds, ligand, instruments or methods should be defined in the text before using throughout the manuscript.

   Examples:
   "The complex cation $Ru(bpy)_3^{2+}$, where bpy = 2,2′-bipyridine, is luminescent . . ."
   "Surfactants such as sodium dodecyl sulfate (SDS) form micelles . . ."
   "Chelate structures were minimized using the empirical force field (EFF) method."
   "Results were analyzed using both molecular orbital (MO) and ligand field (LF) theories"

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6. *Organic abbreviations*—standard organic abbreviations can be used in text and formulae.

   Examples:

   Me = methyl

   Et = ethyl

   iPr = *iso*-propyl

   tBu = *tert*-butyl

   Ch = cyclohexyl

7. *Reagents and solvents*—use chemical formulae for standard reagents and solvents, but not when the name is shorter or more precise.

   Examples: NaOH (*aq*) in place of "sodium hydroxide"

   H_2SO_4 (*aq*) in place of "sulfuric acid"

   CH_2Cl_2 in place of "dichloromethane"

   "caffeine" in place of $C_8H_{10}N_4O_2$

Chemical Terms and Expressions

1. *Chemical names*—the names of chemicals are not capitalized, unless they are trade names such as "Tylenol" or "Viagra."

   incorrect: The reaction of aqueous Cobalt(II) with Aspirin was investigated.

   correct: The reaction of aqueous cobalt(II) with aspirin was investigated.

2. *Create*—chemistry involves "synthesizing" new compounds, "preparing" solutions,7 "characterizing" products. Avoid using phrases such as "products were *created*."

3. *Measurements*—spectra are measured "with" or "using" a spectrometer, not "on" a spectrometer.

4. *Machines*—spectrometers (UV–vis, IR, NMR, etc.) are "instruments," not "machines."

5. *React*—as an intransitive verb, "react" should not have an object and should not have a passive voice. Chemical regents react with each other, they are not reacted.

   incorrect: Hydroxyquinoline and sodium hydroxide were reacted to form sodium hydroxyquinolinate.

   correct: The reaction of hydroxyquinoline and sodium hydroxide produced sodium hydroxyquinolinate.

6. *Tested*—a hypothesis can be "tested" and a student can be "tested." For most laboratory work, the terms "measured," "investigated," "determined," "calculated", or "obtained" often work better.

   incorrect: The absorbance of the solution was tested using the UV–vis machine.

   correct: The absorbance of the solution was measured using a UV–vis spectrophotometer.

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REFERENCES

There are numerous styles for formatting references. Unless otherwise instructed, citations should be formatted in the *JACS* style and appear as endnotes. Alternatively, article titles can also be included. Most important is to prepare citations with a uniform style.

Last name, initials, Last name, initials. *Journal Title* **year**, *volume (issue)*, starting page.
or
Last name, initials, Last name, initials. "Article Title" *Journal Title* **year**, *volume (issue)*, starting page.

Examples:

Schlabach, M., Limbach, H.-H., Shu, A., Bunnenberg, E., Tolf, B., Djerassi, C. *J. Am. Chem. Soc.* **1993**, *115*, 4554.

Additional Materials for Writing Lab/Research Reports

1. Davis, M. *Scientific Papers and Presentations* Academic Press: San Diego, CA, 1997.
2. Dodd, J. S. (ed.). *The ACS Style Guide: a Manual for Authors and Editors*. ACS: Washington, DC, 1997.
3. Eisenberg, A. "Strategies five productive chemists use to handle the writing process." *J. Chem. Educ*. **1982**, *59*, 566.
4. Potera, C. "The Basic Elements of Writing a Scientific Paper: The Art of Scientific Style." *J. Chem. Educ*. **1984**, *61*, 247.
5. Spector, T. "Writing a Scientific Manuscript: Highlights for Success." *J. Chem. Educ*. **1994**, *71*, 47.

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